



TECHNICAL MAINTENANCE MANUAL

Generator Maintenance Manual - UH-GEN-203

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Client	UrbanHive Facility Technologies Pvt. Ltd.
Site	GreenPark Meridian Residency, Whitefield, Bengaluru
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Review cycle	Annual or after any major incident

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1. Equipment Overview and Technical Specifications

Diesel standby generator for common-area backup at GreenPark Meridian. Rated 250 kVA, water-cooled 6-cylinder engine, electronic governor, AMF panel integration with society ATS, and acoustic enclosure for rooftop installation.

Model code: UH-GEN-203

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 415V 3-phase output, 0.8 power factor, neutral earthing ≤ 1 ohm, fuel day tank 8 hours at 70% load

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-GEN-203, commissioning checklist CC-012, and spare parts catalogue SP-UH-GEN-203.

Technicians must verify nameplate data matches AMIS asset record UH-GEN-203 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

2. Installation Guidelines

Install on anti-vibration pads with 150mm clearance for airflow. Exhaust routing per KSPCB norms. Fuel line leak test at 4 bar for 30 minutes. AMF sync test with utility incomer under UrbanHive supervision.

Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD $< 5\%$ on dedicated feeder.
- Confirm earthing resistance ≤ 1 ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily (when on standby): visual leak check, battery float voltage, fuel level log.
- Weekly: idle exercise run 15 minutes; check coolant and oil levels.
- Monthly: load bank test at 30% for 30 minutes; alternator winding IR test.
- Quarterly: fuel filtration, injector spray pattern check, control panel thermography.
- Annual: overhaul per OEM hour meter; replace coolant, filters, and belts as bulletin.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

Fault 01 - Code G101

Symptom: Crank-no-start on AMF signal

Remediation: Check battery 12.6V under load; verify fuel solenoid click; inspect starter motor amp draw < 600A peak.

Fault 02 - Code G205

Symptom: Low oil pressure shutdown

Remediation: Do not restart until oil level confirmed. Inspect pressure sensor and filter bypass valve.

Fault 03 - Code G312

Symptom: High coolant temperature trip

Remediation: Clean radiator fins; verify fan belt tension; test thermostat opening at 82°C.

Fault 04 - Code G408

Symptom: Frequency hunting 48-52 Hz

Remediation: Governor PID retune; inspect fuel rack linkage stiffness; check speed sensor gap 1.0±0.2mm.

Fault 05 - Code G509

Symptom: Alternator insulation alarm

Remediation: Megger test phase-to-earth; dry windings if < 2 M Ω ; inspect AVR card capacitors.

Fault 06 - Code G601

Symptom: Excessive smoke on load step

Remediation: Check air filter restriction; verify injector timing; analyze fuel cetane rating sample.

Fault 07 - Code G702

Symptom: Fuel leak alarm active

Remediation: Isolate day tank valve; inspect flexible hose at engine manifold; replace crushed ferrule.

Fault 08 - Code G803

Symptom: ATS fails to transfer

Remediation: Test utility and generator interlocks; verify coil voltage on contactor; clean auxiliary contacts.

Fault 09 - Code G904

Symptom: Acoustic enclosure over-temp

Remediation: Confirm ventilation louvers open; check exhaust back-pressure < 5 kPa.

Fault 10 - Code G1001

Symptom: Control panel comms fault to BMS

Remediation: Reseat Modbus RS485 terminators; verify baud 9600 8N1; update firmware UH-GEN-AMF-1.4.

Fault 11 - Code G1105

Symptom: Vibration trip on running set

Remediation: Check engine mount rubber pads; inspect coupling alignment within 0.05mm.

Fault 12 - Code G1202

Symptom: Black exhaust after overnight rest

Remediation: Bleed fuel system; test glow plugs; verify compression on suspect cylinder.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

Part No.	Description	Min stock
SP-G01	Oil filter cartridge	8
SP-G02	Fuel filter element	8
SP-G03	Air filter primary	6
SP-G04	Fan belt set	4
SP-G05	AVR control module	1
SP-G06	Coolant thermostat	3

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Hot surfaces and rotating parts during operation. No smoking within 10m of day tank. Fire extinguisher CO2 rated 6kg adjacent to enclosure. Lock-out fuel and battery before maintenance. Report fuel spills to UrbanHive EHS desk immediately.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0012 (08:00-20:00 IST)

Email: support-uh-gen-203@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Generator` with asset_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-012